

New types of smooth subdivision algorithms

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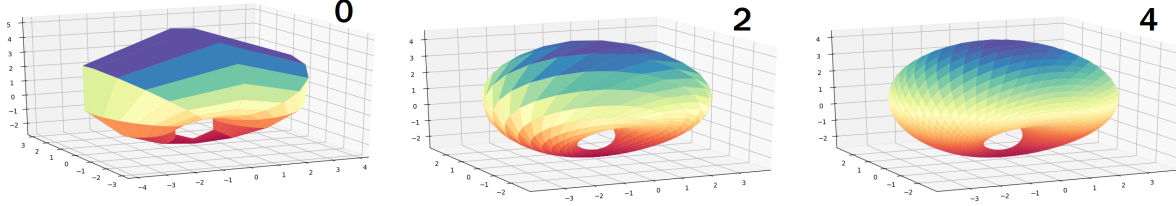


Figure 1: The result after several iterations of the proposed scheme B_3 .

ABSTRACT

We suggest a new type of subdivision schemes based on matrix dilation for generating smooth surfaces. At each iteration, the number of the nodes in the mesh is doubled and the direction of their weighted averaging changes. The scheme has a low complexity because of a small number of coefficients (four, five or six). Using the recent techniques related to the notion of joint spectral characteristics of matrices, we find the smoothness of generated surfaces which in some cases is surprisingly better than for classical schemes.

CCS CONCEPTS

• **Computing methodologies** → *Mesh geometry models*; • **Mathematics of computing** → *Approximation*.

KEYWORDS

subdivision schemes, B-splines, self-similar tiles, wavelets

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Introduction. Subdivision schemes are popular tool for curves and surface modeling. Applying them, we start from a rough mesh (control polygon), and at each iteration we add new nodes and faces using weighted averages of the old nodes. If the scheme converges, we obtain a limit surface for which some properties are desirable. There are many different types of subdivision schemes, for example, stationary (with fixed coefficients of averaging on each iteration) /

non-stationary, acting on different meshes (quad, triangular, etc.), interpolating / extrapolating, etc. We consider the stationary subdivision schemes on regular quad meshes with matrix dilation. First recall how they are defined in the classical setting.

Univariate case. One-dimensional schemes are defined by a finite set of the coefficients $\{c_k\}$. A corresponding subdivision operator S (the symbol) maps a sequence $u \in \ell_\infty(\mathbb{Z})$ to the sequence $Su \in \ell_\infty(\mathbb{Z})$:

$$Su(k) = \sum_{j \in \mathbb{Z}} c_{k-2j} \cdot u(j).$$

Let $f_0(\cdot) = u(\cdot)$ be an initial function ($\mathbb{Z} \rightarrow \mathbb{R}$), then we can define a new function $f_1 : \mathbb{Z}/2 \rightarrow \mathbb{R}$ as $f_1(\frac{1}{2}k) = Su(k)$. Thus, each even value $Su(2k)$ corresponds to the integer point k , $f_1(k) = Su(2k)$, each odd value $Su(2k+1)$ corresponds to the point $k + \frac{1}{2}$, $f_1(k + \frac{1}{2}) = Su(2k+1)$ (see fig. 2). Thus, after one iteration we have a sequence f_1 defined on $\mathbb{Z}/2$. Repeating this process we get sequences f_i defined on $\mathbb{Z}/2^n$ which in case of the convergence tend to a limit curve.

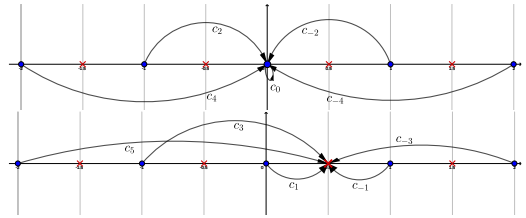


Figure 2: Subdivision rules in univariate case. First: $f_1(0) = Su(0) = \dots + c_2u_{-1} + c_0u_0 + c_{-2}u(1) + \dots$ Second: $f_1(0.5) = Su(1) = \dots + c_1u_0 + c_{-1}u_1 + c_{-3}u(2) + \dots$

The classical multivariate subdivision schemes. A bivariate scheme with coefficients $c_{(k_1, k_2)}$ is given by

$$Su((k_1, k_2)) = \sum_{((j_1, j_2)) \in \mathbb{Z}^2} c_{(k_1-2j_1, k_2-2j_2)} \cdot u(j_1, j_2),$$

where $u = f_0$ is a bivariate sequence $\ell_\infty(\mathbb{Z}^2)$. After one iteration we get values at points $\frac{1}{2}\mathbb{Z}^2$, $f_1(\frac{1}{2}(x, y)) = Su((x, y))$, etc. These rules

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could be rewritten as four independent rules

$$\begin{aligned} Su((2k_1, 2k_2)) &= \sum_{(j_1, j_2)} c_{(2k_1-2j_1, 2k_2-2j_2)} u(j_1, j_2); \\ Su((2k_1 + 1, 2k_2)) &= \sum_{(j_1, j_2)} c_{(2k_1+1-2j_1, 2k_2-2j_2)} u(j_1, j_2); \\ &\dots \end{aligned}$$

where the first rule exploits only coefficients c with even indices, etc. In a particular case, when a bivariate scheme is a direct product of two one-dimensional schemes with coefficients c_k^1, c_k^2 , the operator S is given by

$$Su((k_1, k_2)) = \sum_{(j_1, j_2) \in \mathbb{Z}^2} c_{k_1-2j_1}^1 c_{k_2-2j_2}^2 \cdot u(j_1, j_2).$$

The schemes with matrix dilation. The schemes we consider are based on a dilation integer matrix M , i.e., the matrix with integer entries and all eigenvalues $|\lambda_j| > 1$ (see, for example, [Cavaretta et al. 1991], [Lagarias and Wang 1997]). Let $\{c_k\}_{k \in \mathbb{Z}^2}$ be a set of coefficients, then we define

$$Su(k) = \sum_{j \in \mathbb{Z}^2} c_{k-Mj} \cdot u(j).$$

We restrict our analysis to the matrices M with the property $|\det M| = 2$. It is known that there are only three types of such expanding integer matrices up to affine similarity:

$$M_B = \begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix}, M_D = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, M_R = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}.$$

We introduce a new type of bivariate subdivision scheme based on the matrix M_B , producing smooth limit surfaces and making an irrational rotation of the mesh. The coefficients of the scheme are defined in a special way $c(k_1, 0) = \frac{1}{2^{k_1-1}} \binom{K}{k_1}$ for $k_1 = 0, 1, \dots, K$, which stems from the theory of B-splines. Of our main interest is the matrix M_B since it turn out to produce the limit surface of high regularity. Our main result is as follows:

THEOREM 0.1. *The subdivision scheme B_3 (with $K = 4$) produces C^3 surfaces, the scheme B_2 (with $K = 3$) produces C^2 surfaces, the scheme B_4 (with $K = 5$) produces C^4 surfaces. The classical subdivision schemes of corresponding order do not possess this property.*

Its proof is based on the matrix method from [Charina and Protasov 2019]. It exploits the calculation of joint spectral radius for special matrices [Meystrik 2020].

The example of the scheme B_3 applied for a surface is shown on fig. 1. Let us discuss in details how this scheme works.

In the first iteration we have the values at integer points $\mathbb{Z}^2$. After one iteration, we get the values on $M_B^{-1}\mathbb{Z}^2 = \begin{pmatrix} 0 & 1 \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \mathbb{Z}^2$. The coefficients are $c_{(0,0)} = \frac{1}{8}$, $c_{(1,0)} = \frac{1}{2}$, $c_{(2,0)} = \frac{3}{4}$, $c_{(3,0)} = \frac{1}{2}$, $c_{(4,0)} = \frac{1}{8}$. The rules are as follows (see fig. 3):

$$f_1((0, 0)) = Su((0, 0)) = 0.125u_{(0,0)} + 0.75u_{(0,1)} + 0.125u_{(0,2)};$$

$$f_1((0, 0.5)) = f_1(M_B^{-1}(-1, 0)) = Su((-1, 0)) = 0.5u_{(0,1)} + 0.5u_{(0,2)}.$$

Note that unlike the classical setting, we use only two rules and the number of nodes doubles with each iteration. B_3 is an extrapolating scheme.

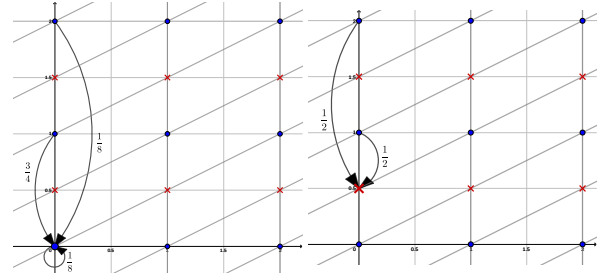


Figure 3: Two types of rules after first iteration of the scheme B_3 .

Most interesting is that happens on next iterations, since the direction of our calculation changes:

$$\begin{aligned} f_2((0, 0)) &= S^2 u((0, 0)) = 0.125Su_{(0,0)} + 0.75Su_{(0,1)} + 0.125Su_{(0,2)} = \\ &= 0.125f_1((0, 0)) + 0.75f_1((1, 0.5)) + 0.125f_1((2, 1)); \end{aligned}$$

$$\begin{aligned} f_2((0.5, 0.25)) &= S^2 u((-1, 0)) = \\ &= 0.5Su_{(0,1)} + 0.5Su_{(0,2)} = 0.5f_1((1, 0.5)) + 0.5f_1((2, 1)). \end{aligned}$$

The coefficients of averaging are same on each iteration, but we multiply the direction by the matrix $M_B^{-1} = \begin{pmatrix} 0 & 1 \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$. As this matrix has the irrational rotation angle with respect to 2π , the directions do not repeat.

Discussion. The advantages of the suggested schemes are a low number of coefficients and high regularity of the resulting surfaces. The key disadvantage is that the scheme is currently applied only for regular meshes, and the generalization is not straightforward.

The coefficients of the suggested schemes stem from the generalization of cardinal B-splines based on special self-similar compacts. This construction is also related to multivariate wavelets [Lagarias and Wang 1997]. There remains an open question on the asymptotic behaviour of smoothness of schemes when K tends to infinity.

There are other generalizations of splines that produce schemes with high regularity such as [Velho and Zorin 2001], [Deng and Ma 2013], and a detailed comparison is a future work.

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